

11

Credit Risk in Energy Markets

11.1 Introduction

Credit Risk is the uncertainty in the value of a portfolio due to the fact that the counterparties to the contracts in a portfolio may be unable to meet their financial obligations. This uncertainty is usually summarised in terms of a discrete index – the credit rating, which can be said to reflect the credit quality of the institution – i.e. the likelihood of the counterparties meeting their contractual obligations.

A major problem with measuring credit risk is the lack of comprehensive credit quality and default data. Whereas VaR methodologies are an exercise in extracting the key properties of market risk from reasonably abundant data, the lack of credit risk data means that measuring credit risk is more an exercise in finding relatively simple models which explain the few key observables.

The measurement and management of credit risk is one of the major challenges facing energy companies today. It is not possible to cover every detail of this complex subject in a single chapter, indeed one could fill many books on this subject alone. It is therefore our goal in this chapter to describe the key concepts and tools in sufficient detail that the reader can make intelligent choices regarding the type of approach which best suits their needs and implement a basic credit risk management system.

11.2 What is Credit Risk?

In general terms, credit risk is the risk of a loss in the value of the portfolio of instruments held due to changes in the credit quality of the counterparties to the instruments. Credit quality measures the probability that the counterparty will be able to meet its financial obligations. The lower the probability that the counterparty will be able to pay, the lower the value of the instrument to the holder relative to its value if payment was certain.

One of the major problems with modelling credit risk is that major decreases in credit quality, particularly default, are relatively rare but cause very large losses. This leads to the distribution of the portfolio values under credit risk having a shape which is difficult to model. Figure 11.1 illustrates typical market and credit risk portfolio return distributions.

In Value-at-Risk models it is reasonable, at least as a first approximation, to model the underlying variables and the portfolio return distribution as normal. In credit risk modelling the key variables and especially the portfolio return cannot be approximated accurately as normally distributed.

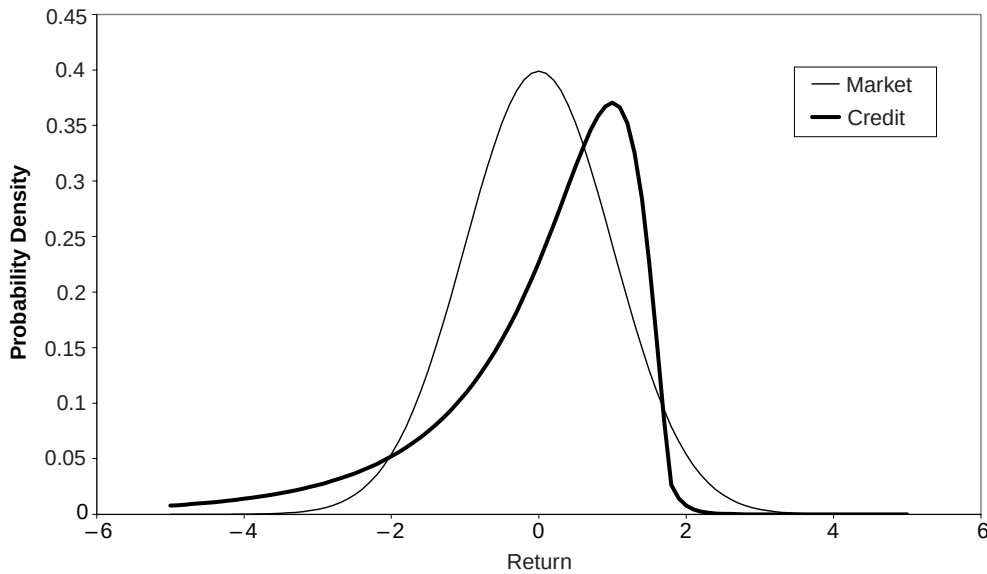


FIGURE 11.1 Typical Market and Credit Risk Distributions

11.3 The Key Components of Credit Risk Models

Credit risk can be thought of as having two main components:

Credit quality risk: This is the risk due to the uncertainty in the future credit quality of the counterparties to the portfolio, and is reflected in the fact that the credit rating of an institution is not fixed but changes depending on its performance in changing economic conditions. Since changes in credit quality imply changes in the likelihood of the company meeting its obligations, this impacts on the value of the instruments held with the counterparty.

Default risk: This is the risk that an institution may actually default on its obligations resulting in a relatively large loss for the counterparty.

In the following sections we discuss the basic observables and data which feed into the majority of credit risk models.

11.3.1 Credit Ratings and Credit Rating Transition Probabilities

A credit rating is a discrete “value” on a pre-defined scale which summarises, in the opinion of the organisation calculating the rating, the overall ability (and willingness) of an institution to meet its financial obligations (i.e. its creditworthiness). Many organisations and companies publish or sell credit ratings of banks, corporations, and other

TABLE 11.1 Moody's Investors Services and Standard and Poor's Credit Rating Scales

	Moody 18	Moody 8	S&P 18	S&P 8
1	Aaa	Aaa	AAA	AAA
2	Aa1	Aa	AA +	AA
3	Aa2	A	AA	A
4	Aa3	Baa	AA–	BBB
5	A1	Ba	A +	BB
6	A2	B	A	B
7	A3	Caa	A–	CCC
8	Baa1	D	BBB +	D
9	Baa2		BBB	
10	Baa3		BBB–	
11	Ba1		BB +	
12	Ba2		BB	
13	Ba3		BB–	
14	B1		B +	
15	B2		B	
16	B3		B–	
17	Caa		CCC	
18	D		D	

financial entities and here we use Moody's Investors Service (Moody) and Standard and Poor's (S&P) credit ratings as examples. Both Moody and S&P provide credit ratings on 18 level and 8 level scales. Table 11.1 shows these four rating scales with the labels which are assigned to each level.

Rating level 1 represents the highest level of creditworthiness on each scale and the credit worthiness decreases down the scale to the lowest level which is default. To give an idea of the relative meaning of the various levels we present a summary of the description of the major Moody's rating levels:

Issuers rated Aaa offer exceptional financial security. While the creditworthiness of these entities is likely to change, such changes as can be visualized are most unlikely to impair their fundamentally strong position.

Issuers rated Aa offer excellent financial security. Together with the Aaa group, they constitute what are generally known as high grade entities. They are rated lower than Aaa entities because long-term risks appear somewhat larger.

Issuers rated A offer good financial security. However elements may be present which suggest a susceptibility to impairment sometime in the future.

Issuers rated Baa offer adequate financial security. However, certain protective elements may be lacking or may be unreliable over any great period of time.

Issuers rated Ba offer questionable financial security. Often the ability of these entities to meet obligations may be moderate and not well safeguarded in the future.

Issuers rated B offer poor financial security. Assurance of payment of obligations over any long period of time is small.

Issuers rated Caa offer very poor financial security. They may be in default on their obligations or there may be present elements of danger with respect to punctual payment of obligations.

Issuers rated Ca offer extremely poor financial security. Such entities are often in default on their obligations or have other marked shortcomings.

Issuers rated C are the lowest rated class of entity, are usually in default on their obligations, and potential recovery values are low.

Credit ratings are assigned after a detailed analysis of the company but many energy companies are not yet rated. Individual energy companies must therefore calculate their own internal ratings. It is generally not feasible for a company to do the depth of analysis performed by the rating organisations, but the general approach is still valid and can be checked by applying the process 'blind' to rated companies and checking the agreement with published ratings.

Associated with each credit rating scale are transition probabilities representing the likelihood that a financial entity with a particular credit rating will retain the same rating or change to a different rating over the next year. The study of credit rating changes is often called migration analysis and many rating organisations also publish or sell this information. This information is obtained from the historical analysis of the changes in ratings in one year of a large set of companies. Credit rating transition probabilities are one of the key inputs in the CreditMetrics model¹, one of the main credit risk models discussed in this chapter, and the information can be obtained from their web site². Table 11.2 shows a typical credit rating transition probability matrix for the S&P 8 level scale.

For a given column each row indicates the probability over a one year period that a company with the credit rating at the top of the column will change to the rating given in the first column. For example, a company with a rating of AA (second column) has a probability of 0.0031 (0.31%) of changing to BBB over the next year.

Notice that for each rating by far the most likely outcome as indicated by the transition probability is for the rating to remain the same. The next most likely event is a single level upgrade or downgrade and multiple upgrades are very unlikely although not impossible. Careful study of table 11.2 reveals its origin in historical analysis via minor anomalies such as the zero probability of a CCC rated company becoming AA rated but having 0.21% chance of becoming AAA rated.

¹ CreditMetrics is a registered trademark of J. P. Morgan

² See www.creditmetrics.com under Datasets.

TABLE 11.2 Credit Rating Transition Probability Matrix

To/From	AAA	AA	A	BBB	BB	B	CCC	D
AAA	0.9082	0.0826	0.0074	0.0006	0.0011	0.0000	0.0000	0.0000
AA	0.0065	0.9088	0.0769	0.0058	0.0005	0.0013	0.0002	0.0000
A	0.0008	0.0242	0.9130	0.0523	0.0068	0.0023	0.0001	0.0005
BBB	0.0003	0.0031	0.0587	0.8746	0.0496	0.0108	0.0012	0.0017
BB	0.0002	0.0012	0.0064	0.0771	0.8116	0.0840	0.0098	0.0098
B	0.0000	0.0010	0.0024	0.0045	0.0686	0.8350	0.0392	0.0492
CCC	0.0021	0.0000	0.0041	0.0124	0.0267	0.1170	0.6448	0.1929
D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000

11.3.2 Credit Spreads

Credit spreads are the differences between the interest rates by which the market discounts credit risky cashflows and the interest rates with which the market discounts certain (default free) cashflows. The idea is that if you are promised a particular cashflow at a specified time in the future but you are not certain that it will be delivered you will calculate the present value of that cashflow to be less than if you considered it certain to be delivered. The lower value can be viewed as equivalent to discounting the cashflow at a higher rate than the markets riskless or default free interest rate. Credit spreads are obtained by comparing the interest rates implied by the market prices of credit risky corporate bonds with the interest rates implied by government bonds (which are considered to have no credit risk – at least for developed countries). Credit spreads can also be obtained from the CreditMetrics web site.

11.3.3 Recovery Rates in Default

In the event of a default of a counterparty, a firm will only receive a fraction of the full amount owed by the counterparty, the fraction is termed the recovery rate. The product of the recovery rate and the total amount owed is equal to the amount which the firm recovers as a result of liquidation or restructuring of the defaulting institution. Recovery rates should take account of the priority (seniority) of the instrument relative to the other obligations in the defaulters portfolio and also any kind of collateral or similar sort of security. Historical recovery rate data indicates that there is significant variation in recovery rates across all levels of seniority. Given this uncertainty, it is important that the uncertainty of recovery rates is modelled if possible and that the sensitivity of the model outputs to the recovery rate assumptions is analysed³. We discuss the modelling of recovery rates in section 11.4.4.

³ This is referred to as stress testing and is discussed in section 11.6.4

TABLE 11.3 A Comparison of the Major Credit Risk Models⁴

	CreditMetrics CreditVaR	CreditRisk +	CreditPortfolio View	Credit Monitor
	JP Morgan	CSFP	McKinsey	KMV Corp.
Definition of Credit Risk	Uncertainty in Market Value	Default Losses	Uncertainty in Market Value	Default Losses
Credit Events	Credit Rating Change/Default	Default	Credit Rating Change/Default	Continuous Default Probabilities
Risk Drivers	Asset Values	Expected Default Rates	Macro Economic Factors	Asset Values
Transition Probabilities	Constant	N/A	Driven by Macro Economic Factors	Driven by expected default rates and asset values
Correlation of credit events	Via asset value model	Via correlated default process	Via economic factors	Via asset values factor model
Recovery Rates	Random (Beta)	Loss given default	Random	Random (Beta)
Computation	Analytical/ Simulation	Analytical	Simulation	Analytical

11.3.4 A Summary of Major Credit Risk Models

Although the data described in the previous sections is a key input into any credit risk model, the major approaches use it in quite different ways. Table 11.3 gives a comparison of the major approaches which are commercially available.

In the following sections we describe the CreditMetrics and CreditRisk + approaches in more detail. We focus on these approaches for two main reasons; the implementation details are readily available and the components are generally the easiest to understand, implement, and integrate with VaR calculations.

11.4 CreditMetrics

The objective of CreditMetrics⁵ is to compute the distribution of changes in the portfolio value at a chosen future date (risk horizon, T_h) due to changes in the credit quality

⁴ CreditVaR is a registered trademark of J. P. Morgan, CreditRisk + is a registered trademark of Credit Suisse Financial Products, CreditPortfolioView is a registered trademark of McKinsey and Co., Credit Monitor is a registered trademark of KMV Corporation.

⁵ Further details can be found in the CreditMetrics Technical Document available from the website www.creditmetrics.com.

(including default) of the counterparties to the portfolio. This is achieved by specifying a model of credit rating changes, a method for revaluation of the portfolio under new credit ratings, and a model of the uncertainty in the recovery rate in default. The model for credit rating changes is based on linking a company's market or asset value returns to credit rating changes via historical credit rating transition probabilities. Correlations between the returns can be specified directly or via country and industry sector correlations to which the companies are associated via specified weightings. Using this approach, a distribution of credit outcomes for the portfolio (credit rating changes including default) at the chosen horizon can be computed. Each credit outcome then has an associated estimated change in the value of the portfolio, calculated using the credit spreads associated with the credit ratings or recovery rates in default. In summary the evaluation of the credit risk involves:

Simulation of changes in the value of the counterparty (together with the underlying market variables, as for VaR).

Changes in the value of the counterparty are mapped to credit rating changes.

Credit ratings are mapped to credit spreads which are used together with the simulated market variables to obtain a distribution of changes in the portfolio value.

Figure 11.2 illustrates the CreditMetrics methodology and data requirements graphically.

The data which is supplied on the CreditMetrics website to support the model includes; the country/industry sector correlations, the credit rating transition probabilities (for the Moody's and Standard and Poor's 18 and 8 level credit rating categorisation), and credit spreads for these ratings for standard maturities of 1, 2, 3, 5, 7, 10, 30 years. In the following sections we describe in detail the steps shown in figure 11.2.

11.4.1 Modelling Credit Rating Changes

Credit Rating changes are assumed to be related to counterparty market value returns. Correlations between credit rating changes of different counterparty's are generated via the correlations between their returns. Note also that the model is based on standardised returns, that is returns after normalising all the volatilities to unity since we are only interested in relative returns and their correlations.

Consider a counterparty with a BBB rating on the scale AAA, AA, A, BBB, BB, B, CCC, Default. We assume there are returns levels L_{CR} , over the specified horizon below which the credit rating changes to CR . Figure 11.3 illustrates a probability density for the returns of a counterparty and the associated credit ratings levels.

Assuming that the returns are normally distributed then the probability of a return being between levels L_{CR1} and L_{CR2} ($L_{CR2} > L_{CR1}$) is given by $N(L_{CR2}) - N(L_{CR1})$ where $N(\cdot)$ is the standard normal cumulative distribution function⁶. As we have discussed, the

⁶ We can make alternative and more realistic assumptions regarding the asset return distribution but the mapping between returns and credit ratings may then require numerical methods.

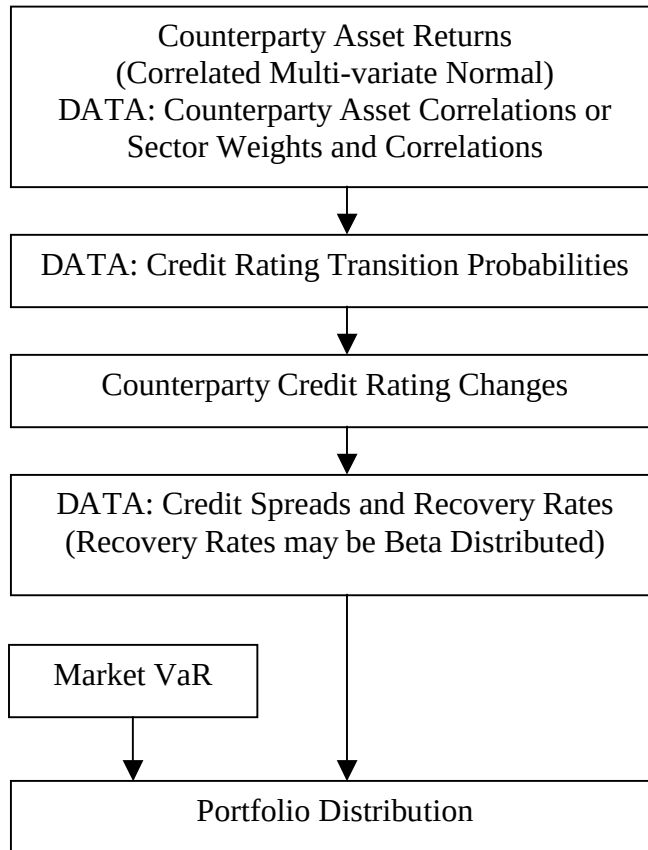


FIGURE 11.2 The CreditMetrics Model

probabilities of transitions from a given credit rating to any other rating can be obtained from the analysis of historical credit rating changes.

Table 11.4 gives an example of the transition probabilities for AAA, AA and BBB rated institutions as of April 1999. By equating the transition probabilities to the probabilities of asset returns between the levels L_{CR} we can solve for the levels which correspond to each credit rating transition. We begin with the default level for which we have;

$$L_{Default} = N^{-1}(\text{Probability}(\text{Default})) \quad (11.1)$$

where $N^{-1}(\cdot)$ is the standard inverse normal distribution function (which is readily available in spreadsheets and libraries of numerical routines). Using $L_{Default}$ we can then

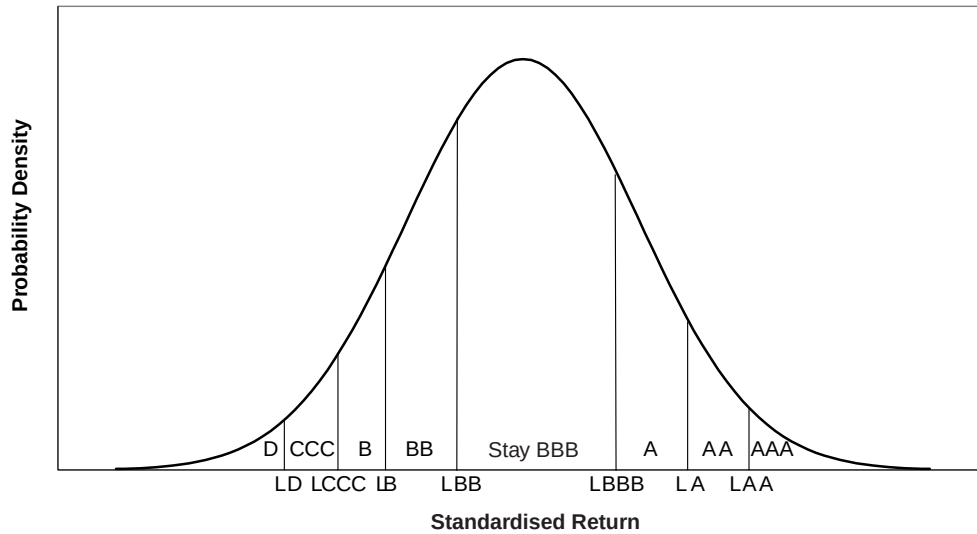


FIGURE 11.3 Counterparty Returns Distribution and Credit Rating Levels

TABLE 11.4 CreditMetrics Credit Rating Transition Probabilities (April 1999)

To / From	AAA	AA	BBB
AAA	0.908214	0.006486	0.000318
AA	0.082649	0.908782	0.003075
A	0.007392	0.076907	0.058736
BBB	0.000616	0.005765	0.874573
BB	0.001129	0.000515	0.049618
B	0	0.001338	0.010814
CCC	0	0.000206	0.001166
Default	0	0	0.001700

obtain L_{CCC} as follows;

$$L_{CCC} = N^{-1}(\text{Probability}(\text{CCC}) + N(L_{\text{Default}})) \quad (11.2)$$

We can continue this process to obtain the levels of all credit rating transitions. This gives us a mapping between standard asset returns and credit ratings. The credit ratings are then mapped to credit spreads (which are available in the CreditMetrics datasets). Table 11.5 gives an example of the mapping between credit ratings, transition probabilities, asset return levels and credit spreads (six month maturity) for a BBB rated institution as of April 1999.

TABLE 11.5 CreditMetrics Credit Rating, Transition Probability, Asset Return Level and Credit Spread Mapping for a BBB Institution (April 1999)

CR	Prob.	L_{CR}	Spread
AAA	0.000318	<i>Not Applicable</i>	0.0040
AA	0.003075	3.415626	0.0050
A	0.058736	2.707166	0.0065
BBB	0.874573	1.537146	0.0093
BB	0.049618	-1.527660	0.0241
B	0.010814	-2.206325	0.0358
CCC	0.001166	-2.762681	0.0500
Default	0.001700	-2.929009	0.05

Note: The spread for the default state is a recovery rate.

We can repeat these calculations for each of the credit ratings of the counterparties in our portfolio. This leads to a mapping between the returns, over the risk horizon T_h , on a counterparty with a particular current credit rating, to their credit rating at the end of the period T_h and an associated credit spread.

11.4.2 Simulation of Counterparty Market Returns

In order to simulate credit rating changes we simulate counterparty market or asset returns and use the model described in the previous section to map from the simulated returns to credit ratings. Let the counterparties to the instruments in the portfolio be denoted by CP_i ; $i = 1, \dots, N_{CP}$ and their asset returns be denoted by x_i ; $i = 1, \dots, N_{CP}$. The returns will, in general, be correlated due in essence to the correlations between the economies of the countries and industry sectors in which the companies are involved. It is these correlations which in turn lead to the correlations between the simulated credit rating changes.

There are two main ways to obtain estimates of the correlations. If we have sufficiently long historical time series of the returns, we can compute the correlations directly from the sample covariance matrix;

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1N_{CP}} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2N_{CP}} \\ \dots & \dots & \dots & \dots \\ \sigma_{N_{CP}1} & \sigma_{N_{CP}2} & \dots & \sigma_{N_{CP}N_{CP}} \end{bmatrix} \quad (11.3)$$

where σ_{ij} denotes the estimate of the covariance between companies i and j . The covariance can be estimated via;

$$\sigma_{ij} = \frac{1}{N} \sum_{k=1}^N (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j) \quad (11.4)$$

where there are N samples ($k = 1, \dots, N$) of x_{ik} and x_{jk} , with sample means $\bar{x}_i, \bar{x}_j$. The correlation matrix $\mathbf{R}$ is given by;

$$\mathbf{R} = \begin{bmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1N_{CP}} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2N_{CP}} \\ \cdots & \cdots & \cdots & \cdots \\ \rho_{N_{CP}1} & \rho_{N_{CP}2} & \cdots & \rho_{N_{CP}N_{CP}} \end{bmatrix} \quad (11.5)$$

where

$$\rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} \quad (11.6)$$

For large numbers of counterparties or where the counterparties do not have easily observable asset values this approach may not be practical. As an alternative, the CreditMetrics datasets include correlations of key country/industry sector indices denoted by $I_j; j = 1, \dots, N_I$. By specifying the weightings (w_{ij}) of the counterparties in the sector indices and the percentage of the counterparty's value which is uncorrelated with the indices (a_i) the correlations between the counterparties can be computed (see CreditMetrics Technical Document, Section 8.5 for more details).

Once we have obtained the correlation matrix we can use the eigenvector decomposition method to simulate the asset returns. Firstly we obtain the eigenvector decomposition of the correlation matrix $\mathbf{R}$;

$$\mathbf{R} = \mathbf{\Gamma} \mathbf{\Lambda} \mathbf{\Gamma}^T \quad (11.7)$$

Where the superscript T denotes the transpose of the matrix and we have;

$$\mathbf{\Gamma} = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1N_{CP}} \\ v_{21} & v_{22} & \cdots & v_{2N_{CP}} \\ \cdots & \cdots & \cdots & \cdots \\ v_{N_{CP}1} & v_{N_{CP}2} & \cdots & v_{N_{CP}N_{CP}} \end{bmatrix} \text{ and } \mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & \cdots & \cdots & \cdots \\ \cdots & \lambda_2 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \lambda_{N_{CP}} \end{bmatrix}$$

The columns of $\mathbf{\Gamma}$ are the eigenvectors. We can then simulate the asset returns x_i at the risk horizon T_h as follows

$$x_{i,T_h} = x_{i,0} + \left(-\frac{1}{2} \sum_{k=1}^{N_{CP}} (\tilde{\sigma}_{ik})^2 \right) T_h + \sum_{k=1}^{N_{CP}} \tilde{\sigma}_{ik} \sqrt{T_h} \varepsilon_k \quad (11.8)$$

where $x_{i,T}$ is the asset return at time T , $\tilde{\sigma}_{ik} = v_{ik} \sqrt{\lambda_k}$, and ε_k is a sample of an independent standard normal random variable. We stress again that we do not need the asset volatilities as we are only interested in relative returns and their correlations which in turn map to correlations between credit ratings. Note that the data and analysis requirements are no more demanding than those that should already be in place to perform the most basic credit risk analysis – i.e. asset market values and their correlations with country and industry sectors.

11.4.3 Credit Spreads, Recovery Rates and the Calculation of Derivative Values

The CreditMetrics specification does not currently describe in detail the method for computing the value of a derivative with a credit risky counterparty. Theoretical analysis by a number of authors (for example; Hull and White (1995), Jarrow and Turnbull (1995), and Longstaff and Schwartz (1995)) have shown that, as long as credit events are uncorrelated with the market, credit risky derivatives can be valued by discounting their expected cashflows by the appropriate credit risky interest rate. It is clear that, in some circumstances, credit events will be correlated with movements in the market. In particular, large downward movements in the market are likely to increase the number of firms getting into financial distress over the following months. However, given the limitations on the data available as a first step the assumption of zero correlation between credit events and market movements gives us a relatively simple solution to the pricing of credit risky derivatives. It is therefore common to assume that the value of a derivative instrument with a credit risky counterparty can be calculated as the value assuming no credit risk discounted at the credit risky interest rate spread.

Denote the value of the instrument with maturity T assuming no credit risk to be C . The counterparty to the instrument has a credit rating of CR and the credit risky interest rate spread for the maturity of the instrument is $Spread_T$. Then the credit risky instrument value C_{CR} is given by

$$C_{CR} = C \exp(-Spread_T T) \quad (11.9)$$

In the case of default, data sources such as CreditMetrics supply Recovery Rates instead of credit spreads. In order to have a consistent data format for calculating the price of instruments, we convert a Recovery Rate (RR) into an effective spread by equating the Recovery Rate with the discount factor implied by an equivalent credit spread $Spread_{Default}$ as follows;

$$RR = \exp(-Spread_{Default} T) \quad (11.10)$$

Rearranging equation (11.10) gives a credit spread which is equivalent to the recovery rate for the purpose of calculating the credit risky value of the instrument;

$$Spread_{Default} = -\ln(RR)/T \quad (11.11)$$

11.4.4 Modelling Recovery Rates

A recovery rate is the fractional amount of the market value of the instrument in the absence of credit risk which is recovered by an institution when the counterparty to the instrument defaults. Research on recovery rates of corporate debt indicates that they are highly uncertain. This uncertainty could potentially contribute significantly to the credit risk of the portfolio and is therefore important to model where possible.

In considering a model for uncertain recovery rates an important factor is that they are bounded between 0 and 1.0. Therefore any distributional model for recovery rates must satisfy this restriction. The classic model for random variables bounded by 0 and 1 is the beta distribution. The beta probability density for RR is given by;

$$f(RR) = 1/B(a,b) RR^{(a-1)} (1-RR)^{(b-1)} \quad (11.12)$$

where $B(a,b)$ is the Beta function. The parameters a and b determine the mean and variance of the distribution and therefore its shape. The mean and variance are given by

$$\text{Mean}[RR] = \mu_{RR} = a/(a+b) \quad (11.13)$$

$$\text{Var}[RR] = \sigma_{RR}^2 = ab/(a+b)^2(a+b+1) \quad (11.14)$$

Equations (11.13) and (11.14) can be rearranged to give a and b in terms of a specified mean μ_{RR} and variance σ_{RR}^2 for the recovery rate;

$$a = (\mu^*/\sigma_{RR}^2 - 1)/(1 + \mu^*) \quad (11.15)$$

$$b = a\mu^*, \mu^* = 1/\mu_{RR} - 1 \quad (11.16)$$

The shape of the beta distribution is very flexible, allowing the most likely recovery rate (the mean) to be set at any level between 0 and 1 and the uncertainty around this value to also be set independently of the mean. Figure 11.4 illustrates some examples of possible recovery rate distributions.

11.4.5 Calculation of Credit Risk for Energy Derivatives

In this section we describe in detail the calculations required to compute the credit risk for a portfolio containing a 1.5 year at-the-money option on a two year oil future and 5×1.5 year far in-the-money options on a two year natural gas future with a valuation date of 30 March 2000. In this example we calculate the credit risk for a one year horizon and we assume that interest rates are flat at 5%. Using our standard notation, with subscripts 1 and 2 denoting the oil and gas contracts respectively, we have the following;

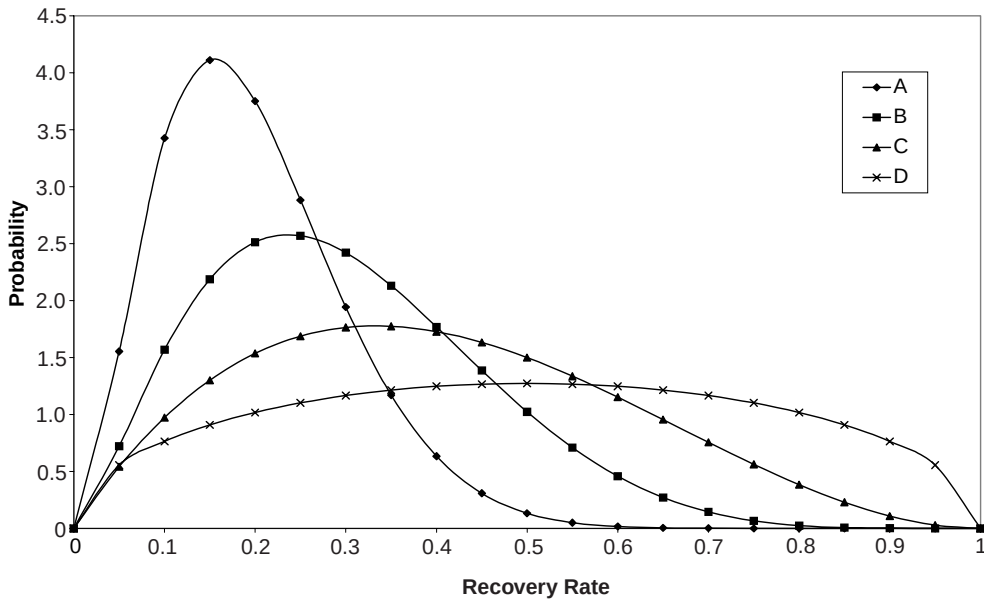
$$K_1 = 21.21, T_1 = 1.5, s_1 = 2.0, F_1 = 22.21$$

$$K_2 = 1.00, T_2 = 1.5, s_2 = 2.0, F_2 = 2.63$$

$$T_h = 1.0$$

In order to price the options and to simulate the underlying forward prices we fit a three factor model of the form described in chapter 8 to two years of historical NYMEX crude oil and natural gas futures prices over the period from 1 April 1998 to 31 March 2000⁷. Table 11.6 shows the current forward curve and volatility functions obtained from the restricted eigenvector decomposition.

⁷ This is the same model as used in chapter 10 for the Value-at-Risk examples.



	A	B	C	D
Mean[RR]	0.2	0.3	0.4	0.5
SD[RR]	0.5	0.5	0.5	0.5
a	3	2.5	2	1.5
b	12	5.833	3	1.5

FIGURE 11.4 Examples of Recovery Rate Distributions

Using the data in Table 11.6 the current prices of the two options are;

$$C_1 = 1.0425, C_2 = 1.5085$$

The two counterparties to the oil and natural gas options (CP1 and CP2 respectively) are assumed to have credit ratings of AAA and BBB and to have a correlation of 75% between their asset market values. Table 11.7 summarises this data.

The first step in the calculation process is to map the counterparty asset returns to credit rating changes and equivalent credit spreads. Recall from sections 11.3.2 and 11.4.3 that for each credit rating the credit spreads vary with the maturity of the instrument considered. In this example both options have 1.5 year maturities and therefore the same credit spread applies to both options. Table 11.8 shows the results of mapping the transition probabilities for the AAA and BBB counterparties to asset return levels and the associated credit spreads.

TABLE 11.6 Three Factor Model Fitted to Historical NYMEX Crude Oil and Natural Gas Futures Price (1 April 1998 to 31 March 2000)

	T	F(0,T)	fn01	fn02	fn03
Oil(1m)	0.0833	26.90	0.5957	−0.0824	0.2160
Oil(2m)	0.1667	26.38	0.4308	−0.0353	−0.1614
Oil(3m)	0.2500	26.04	0.3368	−0.0156	−0.1330
Oil(6m)	0.5000	25.19	0.1641	0.0000	−0.0713
Oil(1y)	1.0000	23.39	0.0948	−0.0017	−0.0371
Oil(2y)	2.0000	21.21	0.0654	0.0021	−0.0270
Gas(1m)	0.0833	2.95	0.0746	0.2827	0.0177
Gas(2m)	0.1667	2.96	0.0647	0.2492	0.0112
Gas(3m)	0.2500	2.97	0.0583	0.2247	0.0087
Gas(6m)	0.5000	2.99	0.0478	0.1846	0.0031
Gas(1y)	1.0000	2.74	0.0397	0.1591	0.0000
Gas(2y)	2.0000	2.63	0.0302	0.1356	−0.0021

TABLE 11.7 Counterparty Credit Ratings and Correlations

		Correlations	
	Credit Rating	CP1	CP2
CP1	AAA	100%	75%
CP2	BBB	75%	100%

CP1 and CP2 indicate counterparty 1 and 2 respectively

TABLE 11.8 Mapping from Counterparty Values to Credit Ratings and Spreads

To/From	AAA	BBB	L _{AAA}	L _{BBB}	Spread
AAA	0.908214	0.000318	N/A	N/A	0.0040
AA	0.082649	0.003075	−1.329836	3.415626	0.0050
A	0.007392	0.058736	−2.359993	2.707166	0.0065
BBB	0.000616	0.874573	−2.920860	1.537146	0.0093
BB	0.001129	0.049618	−3.053865	−1.527660	0.0241
B	0	0.010814	−10	−2.206330	0.0358
CCC	0	0.001166	−10	−2.762680	0.0500
D	0	0.001700	−10	−2.929010	5.9915

In calculating table 11.8 we begin with the AAA counterparty and the default state (D). The transition probability for the AAA rating to ratings D, CCC and B is zero. Theoretically, the result of the inverse cumulative normal applied to zero should be negative infinity. In practice, many implementations of the function will return an error if zero is passed to them. The case of a zero transition probability should therefore be checked and the level L set to large negative number such that the simulated returns have a vanishingly small probability of exceeding that level. We use a level of -10 which is equivalent to a probability of less than 10^{-16} . For the transition to BB we apply equation (11.2) as follows;

$$L_{BB} = N^{-1}(0.01129 + 10^{-16})$$

Which leads to;

$$L_{BB} = -3.053865$$

We proceed in this way, computing the level for each credit rating up to AA. There is no level below which the credit rating changes to AAA because AAA is the highest rating. For the BBB rated counterparty there is a non-zero probability of defaulting over the next year and therefore we first apply equation (11.1) to obtain the default level and then proceed iteratively as before with equation (11.2) to obtain each successive level.

The next step is to simulate the counterparty asset returns and use the mapping we have just computed to convert the simulated returns into simulated credit ratings and associated spreads. Since we only need to simulate two correlated asset returns in this example the eigen decomposition of section 11.4.2 simplifies to the following;

$$x_1 = \varepsilon_1 \quad (11.17)$$

$$x_2 = \rho_{CP} \varepsilon_1 + \sqrt{1 - \rho_{CP}^2} \varepsilon_2 \quad (11.18)$$

where ρ_{CP} is the correlation between the two counterparties and ε_1 and ε_2 are independent standard normal random variables. Similarly, we have only two forward prices to simulate, but we must take into account their volatilities as well as their correlations. The simulation procedure is exactly the same as in chapter 8 and also as in the VaR Monte Carlo simulation in chapter 10;

$$F(T_h, s_k) = F(0, s_k) \exp \left[-\frac{1}{2} \sum_{i=1}^m \{w_{ik}^2 \lambda_i\} + \sum_{i=1}^m \{w_{ik} \sqrt{\lambda_i} \varepsilon_i\} \right] ; \quad k = 1, 2 \quad (11.19)$$

Note that the ε 's in the asset return simulation and forward curve simulation are different. This implies that we are assuming that the asset returns and energy market returns are independent which may not be the case. However, this is a pragmatic approach, estimating correlations between the counterparties asset price returns and energy market prices would be problematic. They would almost certainly be very unstable and therefore likely to simply introduce noise into the risk estimate.

Table 11.9 illustrates the results from the first and last ten simulations (column “sim”) out of 500 of the counterparty asset returns (columns “x1” and “x2”), credit ratings (columns “CR1” and “CR2”) and credit spreads (columns “CS1” and “CS2”). For each simulation, returns “x1” and “x2” are derived from columns “L_{AAA}” and “L_{BBB}” respectively of table 11.8. The row in which the return is above the value in that row but below the value in the row above is selected and the corresponding credit rating and credit spread in that row are returned into table 11.9.

Table 11.10 illustrates the results from the same first and last simulations as table 11.9. The table shows the simulated forward prices (columns “F1” and “F2”), credit spreads (columns “CS1” and “CS2”) as in table 11.9, option prices (columns “C1” and “C2”), portfolio values (column “Port”) and portfolio returns (column “ $\Delta P/P$ ”).

The option prices are computed using the three factor model to compute the credit riskless price and equation (11.9) to adjust these values for the appropriate simulated credit spread. The option price in column “C2” takes into account the fact that five lots of the option are held. The portfolio value is the sum of the values in columns “C1” and “C2” and the portfolio return is the change in the portfolio value divided by its original

TABLE 11.9 Simulated Counterparty Asset Returns, Credit Ratings and Spreads

sim	x1	x2	CR1	CR2	CS1	CS2
1	0.5033	1.3213	AAA	BBB	0.0040	0.0093
2	-1.6192	-1.8709	AA	BB	0.0050	0.0241
3	-0.7027	-0.3615	AAA	BBB	0.0040	0.0093
4	-0.9534	-0.3286	AAA	BBB	0.0040	0.0093
5	0.9758	0.2579	AAA	BBB	0.0040	0.0093
6	-1.5061	-0.7727	AA	BBB	0.0050	0.0093
7	-1.2248	-0.6823	AAA	BBB	0.0040	0.0093
8	-0.8009	-1.7831	AAA	BB	0.0040	0.0241
9	1.1903	0.6361	AAA	BBB	0.0040	0.0093
10	-1.6996	-2.9996	AA	D	0.0050	5.9915
...	...	...	...	...	...	...
491	-0.7682	-0.8794	AAA	BBB	0.0040	0.0093
492	-0.8270	-1.8508	AAA	BB	0.0040	0.0241
493	0.5594	0.2952	AAA	BBB	0.0040	0.0093
494	-1.1936	-1.6348	AAA	BB	0.0040	0.0241
495	-0.6342	0.4204	AAA	BBB	0.0040	0.0093
496	-0.4633	-0.1631	AAA	BBB	0.0040	0.0093
497	1.1435	0.8437	AAA	BBB	0.0040	0.0093
498	-0.3332	0.0623	AAA	BBB	0.0040	0.0093
499	0.3230	0.2139	AAA	BBB	0.0040	0.0093
500	0.1556	-0.8769	AAA	BBB	0.0040	0.0093

TABLE 11.10 Simulated Forward Prices, Credit Spreads, and Portfolio Returns

sim	F1	F2	CS1	CS2	C1	C2	Port	$\Delta P/P$
1	20.859	2.409	0.0040	0.0093	0.6610	6.8391	7.5001	-0.1150
2	19.215	2.543	0.0050	0.0241	0.1697	7.4343	7.6040	-0.1027
3	21.559	2.712	0.0040	0.0093	1.0147	8.3108	9.3255	0.1004
4	19.574	2.330	0.0040	0.0093	0.2399	6.4569	6.6967	-0.2098
5	18.966	2.750	0.0040	0.0093	0.1311	8.4964	8.6276	0.0181
6	22.594	2.526	0.0050	0.0093	1.6904	7.4055	9.0959	0.0733
7	19.254	2.357	0.0040	0.0093	0.1765	6.5857	6.7621	-0.2020
8	19.786	2.319	0.0040	0.0241	0.2902	6.3548	6.6451	-0.2159
9	21.487	3.301	0.0040	0.0093	0.9743	11.1667	12.1410	0.4327
10	20.419	2.992	0.0050	0.0093	0.4843	9.6701	10.1543	0.1983
...	...	...	...	...	...	...	...	...
491	21.395	2.775	0.0040	0.0093	0.9237	8.6151	9.5388	0.1256
492	21.327	2.663	0.0040	0.0241	0.8878	8.0121	8.9000	0.0502
493	21.903	2.768	0.0040	0.0093	1.2209	8.5807	9.8016	0.1566
494	21.818	2.615	0.0040	0.0241	1.1679	7.7813	8.9492	0.0560
495	19.171	2.491	0.0040	0.0093	0.1624	7.2363	7.3987	-0.1269
496	18.959	2.755	0.0040	0.0093	0.1301	8.5182	8.6482	0.0205
497	22.522	3.031	0.0040	0.0093	1.6393	9.8562	11.4954	0.3565
498	19.604	2.901	0.0040	0.0093	0.2466	9.2266	9.4731	0.1179
499	21.151	2.492	0.0040	0.0093	0.7979	7.2432	8.0410	-0.0511
500	20.937	3.454	0.0040	0.0093	0.6957	11.9124	12.6081	0.4878

value. The result of simulating the credit outcomes and market variables at the horizon date is a distribution of the portfolio value. Figure 11.5 shows the histogram of the portfolio returns obtained from the simulations in Table 11.10.

By far the most common occurrence is that of no credit rating changes in which case the changes in the portfolio value are due to market risk alone. Notice the long left tail of decreases in the value of the portfolio due to credit downgrades including a few defaults which give negative returns close to 100%. The long right hand tail is due to the call optionality or leverage of the portfolio. We could calculate the mean and standard deviation of the portfolio value but because of the highly non-normal nature of the distribution the standard deviation will be a poor indicator of the credit risk. A much more useful measure is a percentile level - that is the level below which a chosen percentage of observations fall. Table 11.11 gives typical percentile levels for the distribution in Figure 11.5. For example, the 5th (5%) percentile is the level below which 5% of the observations fall. This indicates that we would expect the portfolio value to fall below this level only once in 20 years. We discuss ways to use outputs from credit risk models in more detail in section 11.6.

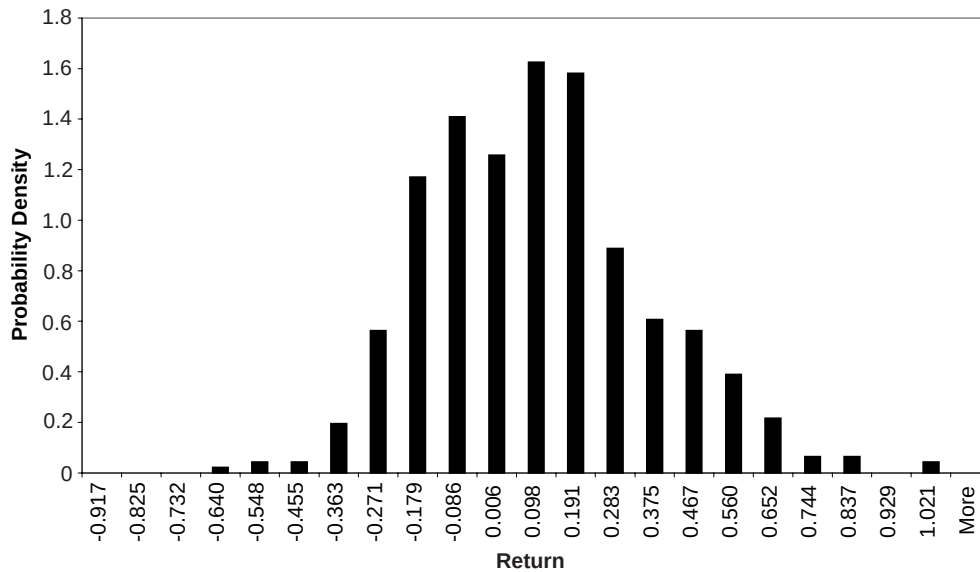


FIGURE 11.5 Typical Portfolio Credit Risk Distribution

TABLE 11.11 Typical Percentile Levels for the Credit Risk Distribution in Figure 11.5

Percentile	Credit Risk Level
1.00%	−49.08%
2.50%	−38.65%
5.00%	−31.45%
10.00%	−24.71%

11.5 CreditRisk +

In the Credit Suisse Financial Products CreditRisk + model, credit risk is considered to be the risk of losses due to the default of counterparties to instruments in the portfolio⁸. The model does not consider the changes in the portfolio value due to changes in the credit quality of the counterparties. One of the main objectives in the construction of the CreditRisk + model was to be able to compute the portfolio loss distribution analytically. The computation of the loss distribution still involves fairly complex calculations involving sums over the relevant variables. The objective of analytical computations

⁸ In the CreditRisk + model counterparties are termed obligors.

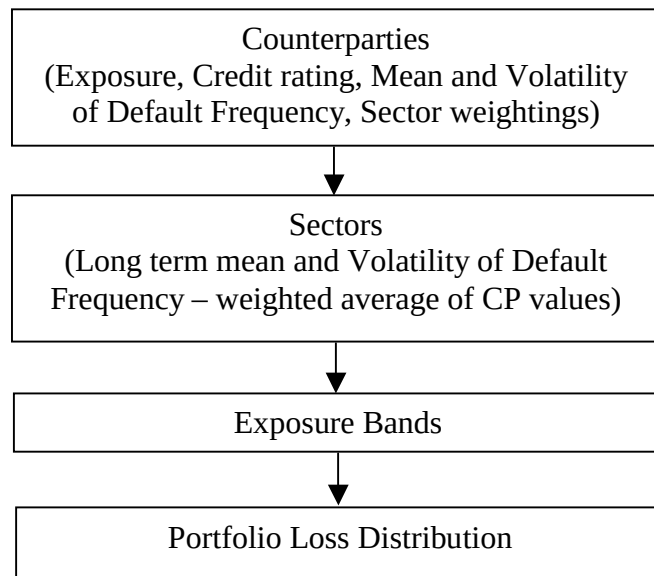


FIGURE 11.6 The CreditRisk+ Model

somewhat restricts the flexibility of the modelling assumptions and we therefore only give an outline of the full model. However, a number of the components of the model provide useful modelling tools in their own right and we discuss these particular components of the CreditRisk+ model in more detail.

The CreditRisk+ model is built around random default rates or frequencies which are assumed to be continuous variables. The variability of default rates is captured by specifying the distribution of default rates and their volatilities with correlations between the losses from different counterparties captured by assigning the counterparties to economic sectors with associated default frequencies and volatilities. Default losses are approximated by integer multiples of a discrete amount, referred to as exposure bands, in order to reduce the computational burden.

The key inputs to the model are: credit exposures, counterparty default rates, counterparty default rate volatilities, and recovery rates. The exposure for each counterparty is the aggregated value of all the deals with that counterparty, taking into account the relevant legal position and any allowable netting. Each counterparty has an average expected default rate and default rate volatility assigned to them. Default rate volatilities are a problematic input to the CreditRisk+ model. They can be obtained from rating agency research publications and typical estimates range from less than 10% for AAA to over 500% for B⁹. However, there is a relatively high degree of uncertainty in these estimates and the results of the model should be analysed for sensitivity to these

⁹ See Carty and Lieberman (1996), Moody's Investors Service Global Credit Research

volatilities. The recovery rate inputs specify the fractional amount of the market value of an instrument which it is estimated would be recovered in the event of the default of a counterparty. Figure 11.6 illustrates the CreditRisk+ model.

Based on the assumptions regarding the structural relationships in the model, it is possible to compute the portfolio loss distribution by essentially summing over the sectors and exposure bands. Note that the CreditRisk+ model, unlike CreditMetrics, does not come supplied with any of the input data and therefore all the input data must be sourced independently by the user. In the following section we describe in detail some of the components of CreditRisk+ which are useful modelling tools in their own right.

11.5.1 Modelling the Occurrence of Defaults

Counterparty defaults are modelled in CreditRisk+ as Poisson events, that is, defaults occur randomly and independently in time with a specified average default rate. Firstly, we will consider the simple case of a constant default rate and then consider the model used in CreditRisk+ with random default rates. The assumption of defaults occurring independently and at random, leads to the following probability density for the number of defaults in a time period T where the average default rate is μ ;

$$\text{Prob}(n) = \exp(-\mu T)(\mu T)^n/n! \quad (11.20)$$

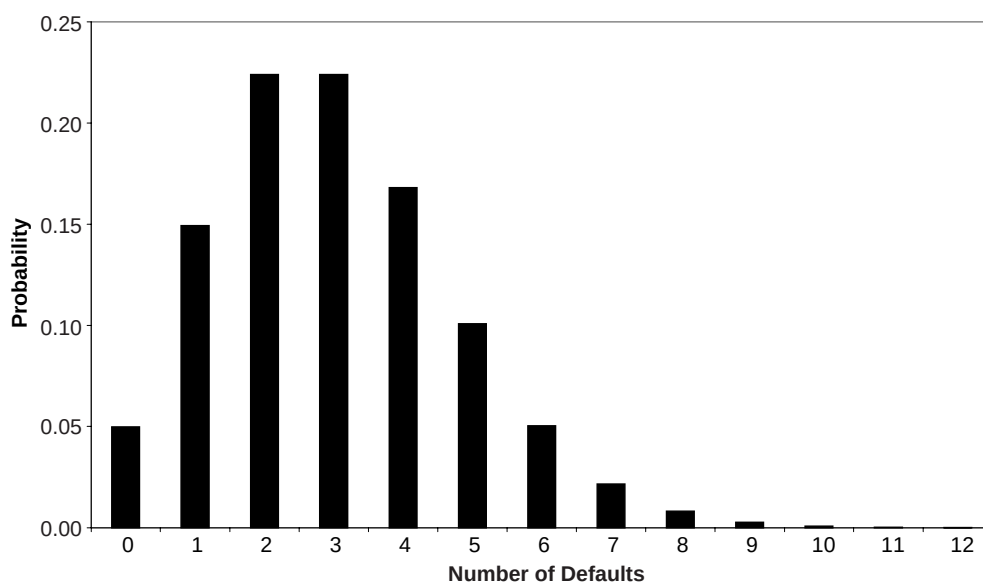


FIGURE 11.7 Distribution of the Number of Defaults (Poisson Default Events). Default frequency $\mu = 3$ per year, Time period $T = 1$ year

The default rate μ is given by the sum of the default probabilities across the counterparties in the portfolio. For example, a portfolio with 100 “B” rated counterparties with a default probability of 3% would give an average default rate of 3 per year, or a portfolio with 10000 “A” rated counterparties with a default probability of 0.03% would also give an average default rate of 3 per year. Figure 11.7 illustrates this probability density for the case of a one year time period and an average default rate of 3 per year.

Given a constant loss per default of, say, \$20,000 then we easily obtain the loss distribution by replacing n in equation (11.20) by the product of n and the loss per default. Figure 11.8 shows the resulting loss distribution.

In reality rates of default in the forthcoming year are highly uncertain and it is important to capture the effect of this on the loss distribution. In CreditRisk+ default rates are modelled as following a Gamma distribution implying the probability density;

$$\text{Prob}(\mu) = 1/(G(a)b^a)(\mu^{a-1})\exp(-\mu) \quad (11.21)$$

$$a = (\mu_l/\sigma_\mu)^2, \quad b = \sigma_\mu^2/\mu_l$$

where μ_l is the long term default rate, σ_μ is the standard deviation of the default rate and $G(\cdot)$ is the gamma function. Figure 11.9 illustrates a typical distribution of the default rates with a long term default rate of 3 per year and a standard deviation of 2 per year.

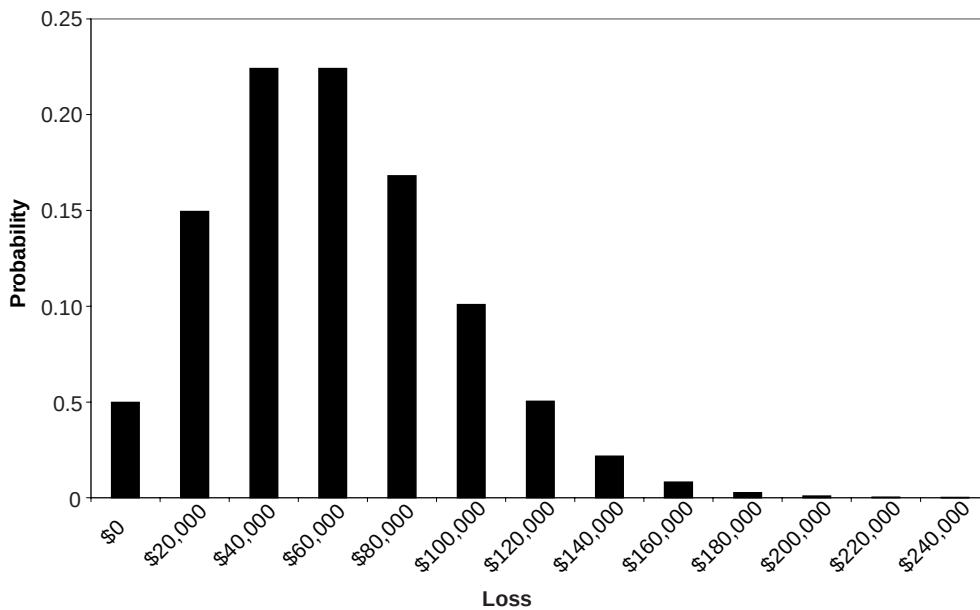


FIGURE 11.8 Example Loss Distribution with Poisson Default Events. Default frequency $\mu = 3$ per year, Time period $T = 1$ year, Loss per Default = \$20,000

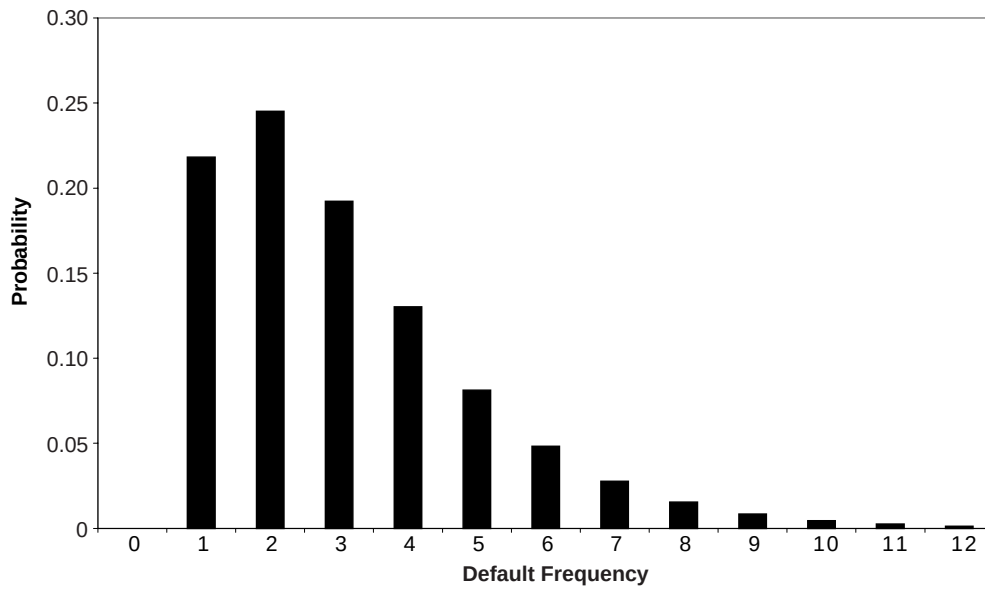


FIGURE 11.9 Distribution of the Default Rate Long Term. Default Rate $\mu_t = 3$ per year, standard deviation $\sigma_\mu = 2$ per year

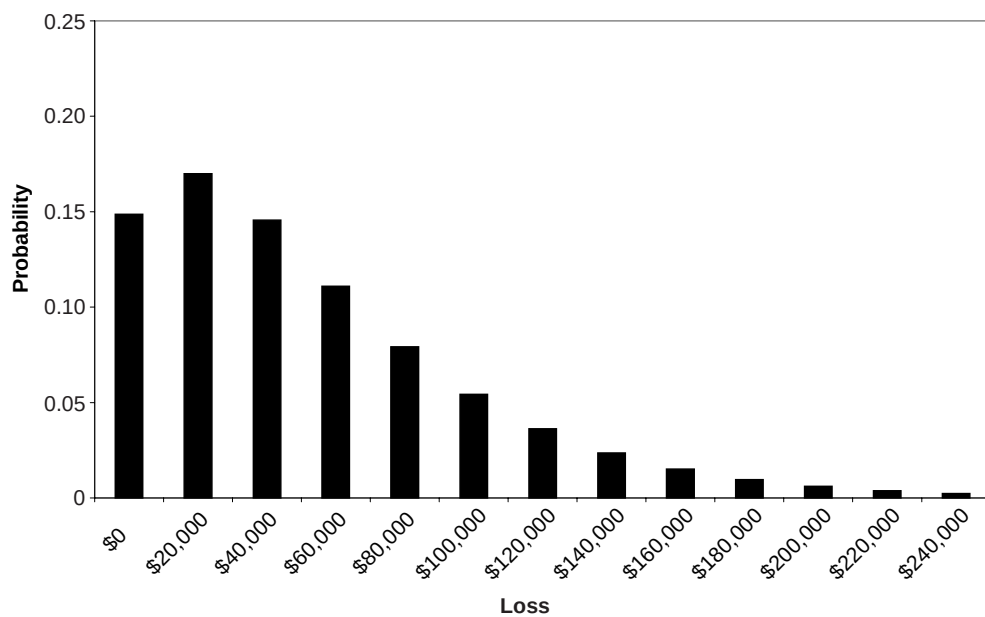


FIGURE 11.10 Example Loss Distribution with Poisson Default Events. Default frequency $\mu = 3$ per year, Time period $T = 1$ year, Loss per Default = \$20,000

The advantage of choosing default rates to be gamma distributed is that the distribution of the number of defaults has the relatively simple analytical form of the negative binomial distribution. Assuming a constant loss per default, the loss distribution is obtained in the same way as before by simply multiplying the number of losses by the constant loss per default. Figure 11.10 shows the resulting loss distribution under the random gamma distributed default rate model with the same \$20,000 loss per default as figure 11.8.

Comparison of figures 11.10 and 11.12 reveal how the random default rate gives a significantly fatter tailed distribution with much higher probabilities of above \$ 150,000.

11.6 Credit Risk Measures and Management

In this section we describe a number of ways in which credit risk models can be used to assess the risk of a portfolio, adjust the make up of the portfolio and set credit risk limits.

11.6.1 Marginal Risk

The marginal risk of a particular instrument is the difference between the risk of the full portfolio including the chosen instrument and the risk of the portfolio with the instrument removed from the portfolio. The risk is usually defined in absolute value terms for a chosen confidence level but can be chosen to suit the users' needs. We illustrate the use of marginal risk using an example portfolio¹⁰. Table 11.12 gives the details of a small example portfolio of twenty instruments with the credit rating of the counterparty (CR), the current market value of the instrument and the marginal risk of the instrument expressed in both absolute dollars and as a percentage of the market value of the instrument.

A useful way to visualise this information is to plot the percentage marginal risk against the market value of the instrument. Figure 11.11 shows the results of plotting the information in table 11.12 together with a line representing a fixed absolute marginal risk of \$60,000.

With the aid of figure 11.11 we can identify the greatest contributors to the portfolio risk. Some of the instruments with the largest absolute risk are easy to interpret. Instrument number 20 is CCC rated and instrument number 11 is BBB rated and has a relatively large exposure. The other "high" risk instruments are less obvious until we consider their correlations. Table 11.13 shows the correlations for the instruments above the solid line in figure 11.11.

Instrument 15 is a reasonably secure BB but it has a relatively high correlation with the B rated instrument and particularly the CCC rated instrument. Similarly, instrument 18 is BB rated but has a 60% correlation with instrument 11 which has a large exposure. The instrument with the highest absolute risk is number 19 which can be attributed to it's correlation with the instruments 15, 18 as well as 20. These instruments are risky because

¹⁰ This example is based on Section 11.4 of the CreditMetrics Technical Document.

TABLE 11.12 Example Portfolio with Marginal Risks

	CR	Market Value	Marginal Risk	
			Absolute	%
1	AAA	\$5,198,807.37	\$222.78	0.00%
2	AAA	\$1,370,613.44	\$112.27	0.01%
3	AA	\$1,091,203.43	\$654.21	0.06%
4	AA	\$13,014,303.76	\$39,171.53	0.30%
5	AA	\$3,869,893.86	\$453.21	0.01%
6	A	\$1,224,195.68	\$2,813.51	0.23%
7	A	\$3,482,327.71	\$4,400.35	0.13%
8	A	\$1,042,809.23	\$1,199.91	0.12%
9	A	\$2,603,423.69	\$4,062.17	0.16%
10	BBB	\$1,100,898.80	\$3,576.36	0.32%
11	BBB	\$10,989,218.64	\$75,555.58	0.69%
12	BBB	\$1,090,782.45	\$128.58	0.01%
13	BB	\$1,082,384.02	\$15,804.62	1.46%
14	BB	\$1,271,977.36	\$42,538.73	3.34%
15	BB	\$4,630,797.85	\$94,456.55	2.04%
16	BB	\$684,195.59	\$31,942.89	4.67%
17	BB	\$1,200,538.80	\$46,509.57	3.87%
18	BB	\$2,285,214.18	\$79,884.47	3.50%
19	B	\$3,073,179.27	\$250,865.13	8.16%
20	CCC	\$1,134,697.24	\$73,395.90	6.47%

of their correlation with the credit risk of the portfolio. Therefore their defaults are likely to be associated with the defaults of other instruments in the portfolio.

11.6.2 Concentration Risk

The discussion in the previous section highlights the fact that one of the major contributors to credit risk is the correlation between defaults amongst instruments in the portfolio. Default correlations are likely to be highest between counterparties which are affected similarly by the factors driving the economy. The economic factors affecting the performance of companies can generally be characterised as being related to the country and industry sector in which the company operates. Companies which operate in the same country and particularly the same industry are likely to have correlated defaults. Therefore, a portfolio which has a large proportion of its instruments with counterparties in the same sector, or indeed with a single counterparty, will tend to have a higher credit risk – this is concentration risk. It is important to be able to capture the

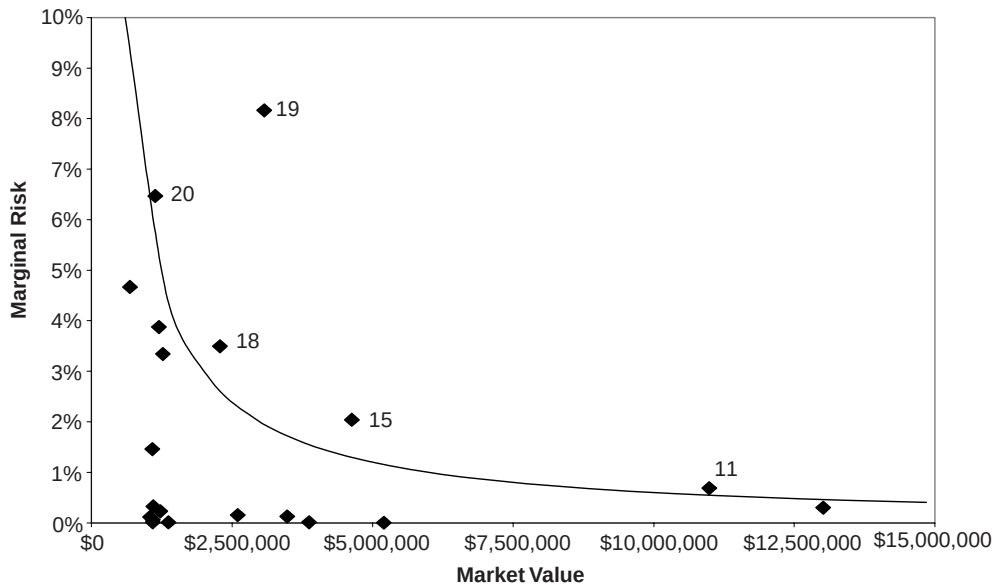


FIGURE 11.11 Percentage Marginal Risk against Market Value of Instrument

TABLE 11.13 Correlations of Most Risky Instruments in Figure 11.11

	11 (BBB)	15 (BB)	18 (BB)	19 (B)	20 (CCC)
11 (BBB)	1.00	0.15	0.60	0.20	0.15
15 (BB)	0.15	1.00	0.15	0.30	0.35
18 (BB)	0.60	0.15	1.00	0.30	0.15
19 (B)	0.20	0.30	0.30	1.00	0.30
20 (CCC)	0.15	0.35	0.15	0.30	1.00

effect of concentration risk in a credit risk model. In CreditMetrics this is done via the model for the asset prices of the counterparties which is driven by sector factors and which in turn drive the credit events, whilst in CreditRisk+ it is sector factors which drive the default rates directly. Concentration risk can be assessed using the marginal risk approach, i.e. by computing the marginal risk of all instruments associated with each counterparty or sector.

11.6.3 Managing Credit Risk

The purpose of the models we have discussed in this chapter is to allow the management of credit risk through the assessment of the size and sources of the credit risk of a

portfolio. The credit risk management processes which are appropriate for a company trading in energy derivatives depends to a large extent on the type of company, the nature of its involvement in energy and the specific energy markets with which it deals. However, it is possible to make some general comments.

Assessment of the risk should ideally be performed using a variety of measures such as:

- Large absolute exposure size – in this case a single default would have large effect on portfolio.

- Large percentage marginal risk – these are most likely to contribute to credit risk associated portfolio losses

Credit risk models can also be used in setting credit risk limits. For example, limits can be set in terms of percentage marginal risk, exposure size, and absolute risk level. Although a credit risk model is not needed to set exposure size limits, it can be used to assess the effect of the limits on the portfolio risk by simulating under different scenarios. One of the main uses of credit risk models is to be able to determine the relative contributions of the portfolio to the overall credit risk so that priorities can be set for actions needed to reduce the risk where it is deemed necessary. Finally, since credit risk models provide the ability to quantitatively measure credit risk they allow the determination of the economic capital required to support the risk of a portfolio.

11.6.4 Assessing the Models

We have already mentioned that credit risk data is sparse and difficult to obtain, thus limiting the accuracy with which the parameters of the models can be estimated. The outputs of any model are obviously determined entirely by the parameter estimates which go into it. It is therefore vitally important to analyse the sensitivity of the model to the parameter values – this is often called “stress testing” or “scenario analysis”. In our experience, parameters which are particularly important to stress test are; recovery rates or recovery rate means and volatilities if random, and asset correlations or sector weights.

The process of stress testing simply involves shifting the chosen parameters up and down by small amounts e.g. 1%, 5%, 10%, 20%, depending the level of uncertainty in the parameter, and recomputing the chosen credit risk statistic e.g. the 95% percentile level. This is obviously a time consuming procedure but it only needs to be as part of the ongoing assessment of the performance of the model and as such might be done for example on only a quarterly basis.

11.7 Conclusions

In this chapter we presented a set of tools for measuring credit risk based on two quite different approaches, *CreditMetrics* and *CreditRisk+* for which detailed public domain

information is available. We also described the data which is publicly available which can be used to specify the parameters of credit risk models. The implementation of the modelling approaches was illustrated with a numerical example using market energy data on a simple energy derivatives portfolio. Finally, we discussed some measures of credit risk which can be computed using the models and general concepts for managing credit risk.